Gesture Detector

13 Gesture Code Reference

Generated from the current gesture_detector codebase on 2026-07-04. This guide documents how each gesture is used in code, which landmarks are read, and which thresholds/angles matter.

Live Detection Priority

- A camera frame is processed only every 100 ms or more.
- Hands below minHandScore 0.45 are ignored for main gesture actions.
- The app first evaluates cancel/OK/call/punch custom gestures and follow-object state.
- Recording holds are processed before zoom and movement.
- Zoom gestures run before generic movement so pinch gestures do not become Moving up.
- Known package gestures such as thumbUp and victory block generic movement.
- Generic direction movement is the fallback action.

Common Geometry Helpers

- handSize = max(hand.boundingBox.width, hand.boundingBox.height).
- palmCenter = average of wrist, indexMCP, middleMCP, ringMCP, and pinkyMCP when available.
- fingerJointAngleDegrees(mcp, pip, tip) measures the PIP angle for non-thumb fingers.
- Visible landmarks require visibility >= 0.35, while zoom uses >= 0.30.
- Image Y grows downward, so upward motion has negative deltaY.

Landmark Point Map Used By This Code

The code uses HandLandmarkType names. The numeric ids below follow the common 21-point hand landmark layout. This project directly references the listed names; id 1 is included only for orientation.

ID	Code point	Where used
0	wrist	Palm center, palm side checks, open palm orientation.
1	thumbCMC	Standard 21-point id; not referenced by this repo.
2	thumbMCP	Thumb angle, closed thumb, open palm thumb score.
3	thumbIP	Thumb angle, call gesture, closed thumb checks.
4	thumbTip	OK touch, call gesture, zoom, open palm, closed thumb.
5	indexMCP	Direction chain, index angle, circle gesture, palm center.
6	indexPIP	Direction chain, index angle, folded/open checks.
7	indexDIP	Direction chain and index circle direction point.
8	indexTip	Direction chain, OK touch, circle tracking, zoom.
9	middleMCP	Direction chain, palm center, open/folded checks.
10	middlePIP	Direction chain and angle/fold checks.
11	middleDIP	Direction chain.
12	middleTip	Direction chain and open/folded checks.
13	ringMCP	Direction chain, palm center, open/folded checks.
14	ringPIP	Direction chain and angle/fold checks.
15	ringDIP	Direction chain.
16	ringTip	Direction chain and open/folded checks.
17	pinkyMCP	Direction chain, palm center, open/folded checks.
18	pinkyPIP	Direction chain and angle/fold checks.
19	pinkyDIP	Direction chain.
20	pinkyTip	Direction chain, call gesture, open/folded checks.

1. Move Left

Move or hold an extended hand/finger shape toward the left. The app reports Moving left.

Code path	• DirectionGestureDetector.detect(...) in lib/hand_gesture_features/domain/services/direction_gesture_detector.dart. • Called from _updateGestureState after custom gestures, recording, zoom, and known package gestures have not won.
How to use in code	• Get bestHand from the camera hand detector. • Call _directionGestureDetector.detect(hand: bestHand, imageSize: detectionImageSize, mirrorHorizontally: mirrorDirectionalGestureCoordinates). • If result == HandMoveDirection.left, show Moving left or send your stand-left command.

Points used	• Index chain: indexFingerMCP -> indexFingerPIP -> indexFingerDIP -> indexFingerTip (ids 5, 6, 7, 8). • Middle chain: middleFingerMCP -> middleFingerPIP -> middleFingerDIP -> middleFingerTip (ids 9, 10, 11, 12). • Ring chain: ringFingerMCP -> ringFingerPIP -> ringFingerDIP -> ringFingerTip (ids 13, 14, 15, 16). • Pinky chain: pinkyMCP -> pinkyPIP -> pinkyDIP -> pinkyTip (ids 17, 18, 19, 20). • X is optionally mirrored before direction math; Y increases downward in image coordinates.
Angles and thresholds	• A finger chain counts only when angle MCP-PIP-TIP is >= 160 deg. • At least 3 accepted finger chains must agree on the same axis. • Horizontal minimum distance = max(imageWidth * 0.025, fingerSpan * 0.18). • Vertical minimum distance = max(imageHeight * 0.025, fingerSpan * 0.18). • Horizontal dominance: abs(deltaX) >= abs(deltaY) * 0.45. • Vertical dominance: abs(deltaY) >= abs(deltaX) * 0.45. • Left condition: summed chain deltaX < 0 after mirroring, and horizontal rules pass.
Overlap priority	• Direction is a fallback. It is skipped while follow-object, recording, zoom, or a known package gesture is active. • Folded fingers no longer count as direction chains, which prevents zoom and fist poses from becoming left movement.

2. Move Right

Move or hold an extended hand/finger shape toward the right. The app reports Moving right.

Code path	• DirectionGestureDetector.detect(...) in direction_gesture_detector.dart. • The same four finger chains are checked as Move Left, but the X direction is positive.
How to use in code	• Call DirectionGestureDetector.detect with the selected hand and detection image size. • If result == HandMoveDirection.right, show Moving right or send your stand-right command. • Keep mirrorHorizontally aligned with the preview/camera direction so left and right are user-visible directions.
Points used	• Index chain: indexFingerMCP -> indexFingerPIP -> indexFingerDIP -> indexFingerTip (ids 5, 6, 7, 8). • Middle chain: middleFingerMCP -> middleFingerPIP -> middleFingerDIP -> middleFingerTip (ids 9, 10, 11, 12). • Ring chain: ringFingerMCP -> ringFingerPIP -> ringFingerDIP -> ringFingerTip (ids 13, 14, 15, 16). • Pinky chain: pinkyMCP -> pinkyPIP -> pinkyDIP -> pinkyTip (ids 17, 18, 19, 20). • X is optionally mirrored before direction math; Y increases downward in image coordinates.
Angles and thresholds	• A finger chain counts only when angle MCP-PIP-TIP is >= 160 deg. • At least 3 accepted finger chains must agree on the same axis. • Horizontal minimum distance = max(imageWidth * 0.025, fingerSpan * 0.18). • Vertical minimum distance = max(imageHeight * 0.025, fingerSpan * 0.18). • Horizontal dominance: abs(deltaX) >= abs(deltaY) * 0.45. • Vertical dominance: abs(deltaY) >= abs(deltaX) * 0.45. • Right condition: summed chain deltaX > 0 after mirroring, and horizontal rules pass.
Overlap priority	• Direction is checked only after higher-priority gestures are rejected. • This stops thumbs-up and zoom from being displayed as movement.

3. Move Up

Point or move an extended hand/finger shape upward. The app reports Moving up.

Code path	• DirectionGestureDetector.detect(...) in direction_gesture_detector.dart. • README also documents pointing up as a Move Up input, but package pointingUp can also be shown as its own known gesture.
How to use in code	• Call DirectionGestureDetector.detect after zoom and package gestures are checked. • If result == HandMoveDirection.up, show Moving up or send your stand-up command.
Points used	• Index chain: indexFingerMCP -> indexFingerPIP -> indexFingerDIP -> indexFingerTip (ids 5, 6, 7, 8). • Middle chain: middleFingerMCP -> middleFingerPIP -> middleFingerDIP -> middleFingerTip (ids 9, 10, 11, 12). • Ring chain: ringFingerMCP -> ringFingerPIP -> ringFingerDIP -> ringFingerTip (ids 13, 14, 15, 16). • Pinky chain: pinkyMCP -> pinkyPIP -> pinkyDIP -> pinkyTip (ids 17, 18, 19, 20). • X is optionally mirrored before direction math; Y increases downward in image coordinates.
Angles and thresholds	• A finger chain counts only when angle MCP-PIP-TIP is >= 160 deg. • At least 3 accepted finger chains must agree on the same axis. • Horizontal minimum distance = max(imageWidth * 0.025, fingerSpan * 0.18). • Vertical minimum distance = max(imageHeight * 0.025, fingerSpan * 0.18). • Horizontal dominance: abs(deltaX) >= abs(deltaY) * 0.45. • Vertical dominance: abs(deltaY) >= abs(deltaX) * 0.45. • Up condition: summed chain deltaY < 0, because image Y gets smaller upward.
Overlap priority	• Zoom is evaluated before movement. If a zoom pose is active, Move Up is skipped. • Folded upward fingers are rejected by the >= 160 deg extension gate.

4. Move Down

Point or move an extended hand/finger shape downward. The app reports Moving down.

Code path	• DirectionGestureDetector.detect(...) in direction_gesture_detector.dart. • The same chain logic as Move Up is used, but the Y direction is positive.
How to use in code	• Call DirectionGestureDetector.detect from the live gesture pipeline. • If result == HandMoveDirection.down, show Moving down or send your stand-down command.
Points used	• Index chain: indexFingerMCP -> indexFingerPIP -> indexFingerDIP -> indexFingerTip (ids 5, 6, 7, 8). • Middle chain: middleFingerMCP -> middleFingerPIP -> middleFingerDIP -> middleFingerTip (ids 9, 10, 11, 12). • Ring chain: ringFingerMCP -> ringFingerPIP -> ringFingerDIP -> ringFingerTip (ids 13, 14, 15, 16). • Pinky chain: pinkyMCP -> pinkyPIP -> pinkyDIP -> pinkyTip (ids 17, 18, 19, 20). • X is optionally mirrored before direction math; Y increases downward in image coordinates.
Angles and thresholds	• A finger chain counts only when angle MCP-PIP-TIP is >= 160 deg. • At least 3 accepted finger chains must agree on the same axis. • Horizontal minimum distance = max(imageWidth * 0.025, fingerSpan * 0.18). • Vertical minimum distance = max(imageHeight * 0.025, fingerSpan * 0.18). • Horizontal dominance: abs(deltaX) >= abs(deltaY) * 0.45. • Vertical dominance: abs(deltaY) >= abs(deltaX) * 0.45. • Down condition: summed chain deltaY > 0, because image Y gets larger downward.
Overlap priority	• Movement is ignored when follow tracking, custom gestures, recording, zoom, or known package gestures are active.

5. Detect My Face

Hold the call-me hand shape for 2 seconds. The app then searches for the best face target.

Code path	• CustomGestureDetector._isCallMeGesture(...) sets rawCustomGestureResult.isCallMe. • _updateGestureState holds _faceDetectGestureStartedAt until faceDetectHoldDuration is reached, then calls _selectBestFaceTarget(...).
How to use in code	• Call _customGestureDetector.detect(...) for the selected hand. • If result.isCallMe is true and no follow target is already locked, start/continue a 2 second hold timer. • When hold progress reaches 1.0, call face detection and lock the best face target.
Points used	• Thumb: thumbTip and thumbIP (ids 4, 3). • Pinky: pinkyTip and pinkyPIP (ids 20, 18). • Closed fingers: indexTip/indexPIP (8, 6), middleTip/middlePIP (12, 10), ringTip/ringPIP (16, 14). • Palm center: average of wrist plus index/middle/ring/pinky MCP points.
Angles and thresholds	• Thumb open: dist(thumbTip, palmCenter) > dist(thumbIP, palmCenter) * 1.15. • Thumb reach: dist(thumbTip, palmCenter) > handSize * 0.23. • Pinky open: tip distance > pip distance * 1.20 and tip distance > handSize * 0.30. • Index, middle, ring closed: tip distance <= pip distance * 1.03 OR tip distance < handSize * 0.26. • Thumb and pinky separation: dist(thumbTip, pinkyTip) > handSize * 0.55. • Face detect hold: 2 seconds.
Overlap priority	• This custom gesture runs before movement and zoom display, so it should not fall through to Moving left/up.

6. Follow The Object

Open palm, hold, closed fist, then open palm again. The final hand-box center is used as the release point.

Code path	• FollowObjectSequenceDetector.update(...) controls the sequence. • Open palm is custom geometry via OpenPalmGestureDetector. Closed fist in this sequence is package GestureType.closedFist. • _selectFollowTargetAtReleasePoint(...) selects a face or object at release.
How to use in code	• Show open palm and hold for followObjectFirstOpenPalmHoldDuration (1 second). • Show closed fist; this moves the sequence to waitingForFinalOpen. • Move the hand over the object/face and open palm again. • Use releasePoint = center of hand.boundingBox to choose the target.
Points used	• Open palm uses wrist plus thumbMCP/thumbIP/thumbTip and index/middle/ring/pinky MCP/PIP/tip points. • Open palm also uses indexTip, middleTip, ringTip, pinkyTip spread distances. • Closed fist for this sequence is package-classified; this repo does not define its sequence points. • Release point is not a landmark; it is the center of the detected hand bounding box.
Angles and thresholds	• Open palm enter confidence: 0.55; exit confidence: 0.45. • Smoothing: 4 samples, at least 2 positive samples, max sample age 500 ms. • Four fingers score angle 145..165 deg, tip/pip distance ratio 1.08..1.22, reach 0.25..0.34 of handSize. • Thumb score angle 125..155 deg plus thumb reach and separation from index. • Spread score uses indexTip to pinkyTip, adjacent tip distances, and indexMCP to pinkyMCP fan ratio. • Closed fist package confidence: >= 0.50 for the follow sequence.
Overlap priority	• While the follow sequence is active, custom gestures, recording, zoom, and movement are suppressed.

7. Stop and Continue Action

Show thumbs-up. The current code displays Stop & Continue Action for a package thumbUp gesture.

Code path	• Package classifier result: hand.gesture.type == GestureType.thumbUp. • Handled in _updateGestureState under hasKnownGesture.
How to use in code	• Read bestHand.gesture after follow tracking and custom gestures are handled. • Require gesture.type == GestureType.thumbUp and confidence >= minPackageGestureConfidence (0.50). • Set gesture text or run your stop/continue command. • README says hold 1 second; current code does not yet add a thumbUp hold timer.
Points used	• No direct landmark math for thumbUp exists in this repo. • The hand_detection package decides thumbUp internally and returns GestureType.thumbUp. • If you want local geometry later, use thumbTip/thumbIP/thumbMCP and folded index/middle/ring/pinky checks.
Angles and thresholds	• Package gesture confidence: >= 0.50. • No repo-defined angle thresholds for thumbUp.
Overlap priority	• Known package gestures now block generic movement, so thumbUp should not show Moving left.

8. Return To Main Position

Rotate the index finger in a small circle while only the index finger is extended upward.

Code path	• CustomGestureDetector._detectCancelEverythingGesture(...). • Result label: Return to main position. In the live screen this clears active tasks and resets camera zoom.
How to use in code	• Call _customGestureDetector.detect(...) every processed frame. • When result.isCancelEverything is true, call _clearAllActiveGestureTasks(resetCameraZoom: true). • Keep passing mirrorHorizontally so index-tip circle points match visible direction.
Points used	• Index-only pose uses thumbTip/thumbIP/thumbMCP, indexMCP/indexPIP/indexDIP/indexTip, middle/ring/pinky MCP/PIP/tip. • Circle history tracks only visible indexFingerTip (id 8). • Palm center uses wrist and four MCP points.
Angles and thresholds	• Index extended: MCP-PIP-TIP angle >= 160 deg and tip distance > handSize * 0.30. • Index faces up: indexTip.y < indexPIP.y and indexTip.y < palmCenter.y - handSize * 0.08. • Index upright side offset: abs(indexTip.x - indexMCP.x) <= handSize * 0.50. • Thumb closed: thumbTip to palm <= 0.30 * handSize; thumbTip to IP ratio <= 1.00. • Thumb close to knuckles: min(thumbTip-indexMCP, thumbTip-middleMCP) <= 0.32 * handSize. • Thumb not stretched: thumbTip-thumbMCP <= 0.36 * handSize. • Middle, ring, pinky folded by angle: <= 145 deg. • Circle history: max 36 samples, 1400 ms window, at least 5 samples. • Circle radius: max(min(imageWidth, imageHeight) * 0.006, handSize * 0.015). • Radius variation: <= averageRadius * 1.60. • Total circular angle: abs(totalAngle) >= pi * 0.90.
Overlap priority	• This custom gesture is handled before recording, zoom, package gestures, and direction movement.

9. Start Record Video

Hold the OK gesture for 1 second. The app starts video recording.

Code path	• CustomGestureDetector._isOkGesture(...) sets customGestureResult.isOk. • recording_controls.dart maps isOk to _RecordingGestureAction.start.
How to use in code	• Call _customGestureDetector.detect(...) and require exactly one custom label. • If result.isOk is true, pass _RecordingGestureAction.start to _updateRecordingGestureHold. • After recordStartHoldDuration reaches 1 second, call _startGestureVideoRecording(controller).
Points used	• Thumb/index touch: thumbTip and indexTip (ids 4, 8). • Index bend: indexMCP, indexPIP, indexTip (5, 6, 8). • Open fingers: middle/ring/pinky MCP/PIP/tip. • Palm center for extension checks.
Angles and thresholds	• Thumb-index touch distance <= max(handSize * 0.11, 12 px). • Index bend angle MCP-PIP-TIP <= 150 deg. • Middle/ring/pinky extended by angle: >= 160 deg and tip distance > handSize * 0.30. • Hold duration: 1 second. • Only allowed when not already recording.
Overlap priority	• Recording gestures run before zoom and movement. • If multiple custom gestures overlap, the app shows Hand detected instead of triggering recording.

10. Pause Video

Make a fist and hold for 1 second. The app toggles pause/resume while recording.

Code path	• CustomGestureDetector._isPunchGesture(...) sets customGestureResult.isPunch. • recording_controls.dart maps isPunch to _RecordingGestureAction.togglePause.
How to use in code	• Call _customGestureDetector.detect(...) and require exactly one custom label. • If result.isPunch is true and video is recording, pass togglePause to _updateRecordingGestureHold. • After 1 second, call pauseVideoRecording() or resumeVideoRecording().
Points used	• Index, middle, ring, pinky tip/PIP/MCP points. • Palm center for folded checks. • Knuckle alignment uses y values of indexMCP, middleMCP, ringMCP, pinkyMCP.
Angles and thresholds	• Each finger folded: tip distance <= pip distance * 1.03 OR tip distance < handSize * 0.26. • Knuckle Y spread <= handSize * 0.12. • Package closedFist support can help when confidence >= 0.60, but fingers still must be closed. • Hold duration: 1 second. • Only allowed while video is already recording.
Overlap priority	• Recording feedback suppresses movement and zoom while the hold is active.

11. End Record Video

Hold the victory gesture for 2 seconds. The app stops and saves the recording.

Code path	• Package classifier result: hand.gesture.type == GestureType.victory. • recording_controls.dart maps hasVictoryGesture to _RecordingGestureAction.stop.
How to use in code	• Ensure follow tracking is not active and no custom gesture label is present. • Require GestureType.victory with confidence >= 0.50. • Pass _RecordingGestureAction.stop to _updateRecordingGestureHold. • After 2 seconds, call _stopGestureVideoRecording(controller).
Points used	• No direct victory landmark math exists in this repo. • The hand_detection package decides victory internally. • Typical victory geometry would involve index/middle extended and ring/pinky folded, but that is not implemented here.
Angles and thresholds	• Package gesture confidence: >= 0.50. • Hold duration: 2 seconds. • Only allowed while video is recording.
Overlap priority	• Victory is checked before zoom/movement and is used only for recording stop.

12. Zoom In

Start with thumb and index pinched, hold briefly, then open them outward.

Code path	• ZoomGestureDetector.detect(...) state machine. • _handleZoomDirection(ZoomDirection.zoomIn) applies camera zoom by +zoomStep.
How to use in code	• Call _zoomGestureDetector.detect(...) after custom/recording checks and before movement. • Start pose: thumb-index distance ratio is closed. • Movement: open thumb/index enough within max gesture duration. • If result == ZoomDirection.zoomIn, call _handleZoomDirection(result).
Points used	• Active pinch points: thumbTip and indexFingerTip (ids 4, 8). • Middle/ring/pinky folded check uses their MCP/PIP/tip points. • Palm center and hand bounding box define handSize and active tip-center distance.
Angles and thresholds	• Middle, ring, pinky folded by angle: <= 145 deg. • Thumb/index distance ratio = dist(thumbTip, indexTip) / handSize. • Valid active tips: center of thumbTip/indexTip must be farther than handSize * 0.06 from palm center. • Closed start ratio: <= 0.26 and held for 500 ms. • Open finish ratio: >= 0.27 and increased by at least 0.035. • Gesture stable duration: >= 90 ms; max gesture duration: 2600 ms. • Detected result is held for 650 ms. Camera zoom step is +0.2.
Overlap priority	• Zoom is evaluated before direction movement, and active zoom blocks Move Up/Down/Left/Right.

13. Zoom Out

Start with thumb and index open, hold briefly, then pinch them together.

Code path	• ZoomGestureDetector.detect(...) state machine. • _handleZoomDirection(ZoomDirection.zoomOut) applies camera zoom by -zoomStep.
How to use in code	• Call _zoomGestureDetector.detect(...) while no higher-priority custom/recording gesture is active. • Start pose: thumb-index distance ratio is open. • Movement: close thumb/index enough within max gesture duration. • If result == ZoomDirection.zoomOut, call _handleZoomDirection(result).
Points used	• Active pinch points: thumbTip and indexFingerTip (ids 4, 8). • Middle/ring/pinky folded check uses their MCP/PIP/tip points. • Partial zoom-out recovery can use only thumbTip/indexTip divided by image max side.
Angles and thresholds	• Open start ratio: >= 0.27 and held for 500 ms. • Closed finish ratio: <= 0.26 and decreased by at least 0.035. • Gesture stable duration: >= 90 ms; max gesture duration: 2600 ms. • Partial zoom-out open image ratio: >= 0.045. • Partial close change: >= 0.018 and distance <= startDistance * 0.72. • Detected result is held for 650 ms. Camera zoom step is -0.2.
Overlap priority	• Zoom out uses the same priority protection as Zoom In and blocks generic movement while active.